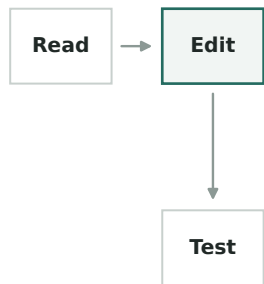


Precedence-constrained attribution under a controlled protocol

controlled structural protocol

(a) Declared precedence DAG

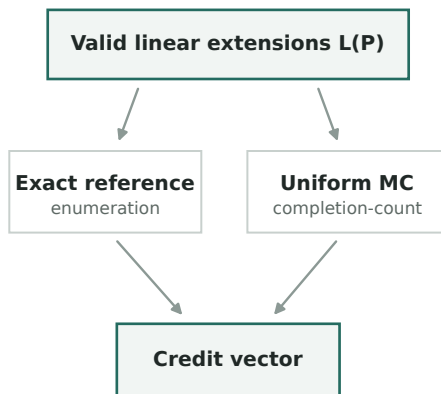


valid: Read -> Edit -> Test

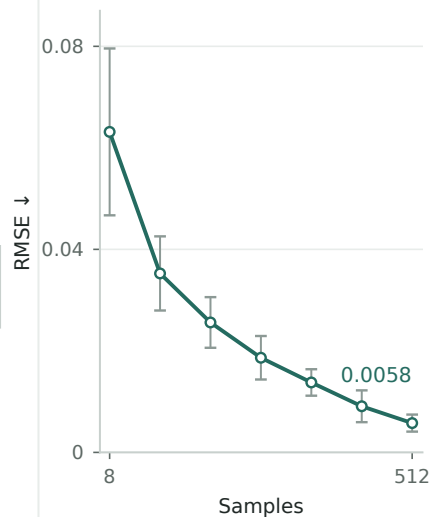
~~excluded: Edit -> Read~~

(b) Ordered-Shapley estimation

$$\phi_i^P = \mathbb{E}_{\pi \in \mathcal{L}(P)} [v(S_i^\pi \cup \{i\}) - v(S_i^\pi)]$$



(c) RMSE convergence



n = 12 seeds

points: mean · bars: reported 95% CI